INSTRUCTION BOOK

GE MEDICAL
3T POWER MEASUREMENT KIT

Per GE Spec No. 2372868PSP
Prior approval required before making any changes that affect form, fit, or function. Release of manual changes required at least 15 days before delivery of changed product. Send 2 hard copies and 1 pdf to:
Manager, MR Service Engineering, W-827
GE Medical Systems
PO Box 414
Milwaukee, WI 53201

Electronic Corporation
Cleveland (Solon) Ohio USA

Instruction Book Part Number 920-4300A670 Rev. C

The following are general safety precautions that are not necessarily related to any specific part or procedure, and do not necessarily appear elsewhere in this publication. These precautions must be thoroughly understood and apply to all phases of operation and maintenance.

Keep Away From Live Circuits

Operating personnel must at all times observe normal safety regulations. Do not replace components or make adjustments inside the equipment with high voltage turned on. To avoid casualties, always remove power.

Shock Hazard

Do not attempt to remove the RF transmission line while RF power is present. Radiated RF power is a potential health hazard.

Do Not Service or Adjust Alone

Under no circumstances should any person reach into an enclosure for the purpose of service or adjustment of equipment except in the presence of someone who is capable of rendering aid.

Chemical Hazard

Dry cleaning solvents for cleaning parts may be potentially dangerous. Avoid inhalation of fumes or prolonged contact with skin.

Safety Earth Ground

An uninterruptible earth safety ground must be supplied from the main power source to test instruments. Grounding one conductor of a two conductor power cable is not sufficient protection. Serious injury or death can occur if this grounding is not properly supplied.

Resuscitation

Personnel working with or near high voltages should be familiar with modern methods of resuscitation.

Safety Symbols

WARNING Warning notes call attention to a procedure which, if not correctly performed, could result in personal injury.

CAUTION Caution notes call attention to a procedure which, if not correctly performed, could result in damage to the instrument.

The caution symbol appears on the equipment indicating there is important information in the instruction manual regarding that particular area.

This symbol indicates that the unit radiates heat and should not be touched while hot.

☞ NOTE: Calls attention to supplemental information

Warning Statements

The following safety warnings appear in the text where there is danger to operating and maintenance personnel, and are repeated here for emphasis.

WARNING Never attempt to connect or disconnect RF equipment from the transmission line while RF power is being applied. Leaking RF energy is a potential health hazard.

WARNING Disconnect from external power before any disassembly. The potential for electric shock exists.

WARNING Load may be hot. Allow load to cool before touching.

Caution Statements

The following equipment cautions appear in the text whenever the equipment is in danger of damage, and are repeated here for emphasis.

CAUTION Use a Bird adapter only and do not use the adapter with the batteries removed.

CAUTION Harsh or abrasive detergents and some solvents can damage the display unit and information on the labels.

CAUTION Always turn off the DPM before connecting or disconnecting a sensor.

CAUTION Replace only with Ni-MH rechargeable A batteries. Nominal Voltage 1.2V; Capacity 2700mAhr.

CAUTION Packing foam may melt if heated. Allow load to cool before packing.

Safety Statements

USAGE

ANY USE OF THIS INSTRUMENT IN A MANNER NOT SPECIFIED BY THE MANUFACTURER MAY IMPAIR THE INSTRUMENT'S SAFETY PROTECTION.

USO

EL USO DE ESTE INSTRUMENTO DE MANERA NO ESPECIFICADA POR EL FABRICANTE, PUEDE ANULAR LA PROTECCIÓN DE SEGURIDAD DEL INSTRUMENTO.

BENUTZUNG

WIRD DAS GERÄT AUF ANDERE WEISE VERWENDET ALS VOM HERSTELLER BESCHRIEBEN, KANN DIE GERÄTESICHERHEIT BEEINTRÄCHTIGT WERDEN.

UTILISATION

TOUTE UTILISATION DE CET INSTRUMENT QUI N'EST PAS EXPLICITEMENT PRÉVUE PAR LE FABRICANT PEUT ENDOMMAGER LE DISPOSITIF DE PROTECTION DE L'INSTRUMENT.

IMPIEGO

QUALORA QUESTO STRUMENTO VENISSE UTILIZZATO IN MODO DIVERSO DA COME SPECIFICATO DAL PRODUTTORE LA PROZIONE DI SICUREZZA POTREBBE VENIRNE COMPROMESSA.

SERVICE

SERVICING INSTRUCTIONS ARE FOR USE BY SERVICE - TRAINED PERSONNEL ONLY. TO AVOID DANGEROUS ELECTRIC SHOCK, DO NOT PERFORM ANY SERVICING UNLESS QUALIFIED TO DO SO.

SERVICIO

LAS INSTRUCCIONES DE SERVICIO SON PARA USO EXCLUSIVO DEL PERSONAL DE SERVICIO CAPACITADO. PARA EVITAR EL PELIGRO DE DESCARGAS ELÉCTRICAS, NO REALICE NINGÚN SERVICIO A MENOS QUE ESTÉ CAPACITADO PARA HACERIO.

WARTUNG

ANWEISUNGEN FÜR DIE WARTUNG DES GERÄTES GELTEN NUR FÜR GESCHULTES FACHPERSONAL.

ZUR VERMEIDUNG GEFÄHRLICHE, ELEKTRISCHE SCHOCKS, SIND WARTUNGSARBEITEN AUSSCHLIEßLICH VON QUALIFIZIERTEM SERVICEPERSONAL DURCHZUFÜHREN.

ENTRENTIEN

L'EMPLOI DES INSTRUCTIONS D'ENTRETIEN DOIT ÊTRE RÉSERVÉ AU PERSONNEL FORMÉ AUX OPÉRATIONS D'ENTRETIEN. POUR PRÉVENIR UN CHOC ÉLECTRIQUE DANGEREUX, NE PAS EFFECTUER D'ENTRETIEN SI L'ON N'A PAS ÉTÉ QUALIFIÉ POUR CE FAIRE.

RF VOLTAGE MAY BE PRESENT IN RF ELEMENT SOCKET - KEEP ELEMENT IN SOCKET DURING OPERATION.

DE LA TENSION H.F. PEAT ÊTRE PRÉSENTE DANS LA PRISE DE L'ÉLÉMENT H.F. - CONSERVER L'ÉLÉMENT DANS LA PRISE LORS DE L'EMPLOI.

HF-SPANNUNG KANN IN DER HF-ELEMENT-BUCHSE ANSTEHEN - ELEMENT WÄHREND DES BETRIEBS EINGESTÖPSELT LASSEN.

PUEDE HABER VOLTAJE RF EN EL ENCHUFE DEL ELEMENTO RF - MANTENGA EL ELEMENTO EN EL ENCHUFE DURANTE LA OPERACION.

IL PORTAELEMENTO RF PUÒ PRESENTARE VOLTAGGIO RF - TENERE L'ELEMENTO NELLA PRESA DURANTE IL FUNZIONAMENTO.

About This Manual

This manual guides users through the operation and maintenance of the GE Medical 3T Power Measurement Kit (GE3TKIT), including the Bird 5000-EX Digital Power Meter (DPM), Bird 5010G Directional Power Sensor (DPS) and Bird 8581A Air-Cooled Load.

Typography

There are two types of keys on the DPM. A hard key has a specific function which is indicated on the key. The key names for hard keys are set in bold typeface, e.g. *Press the* **ON** *key*.

Speed keys, which are under the display, have a different label depending upon the function selected. The names appear at the bottom of the display, directly above the corresponding key. The key names for speed keys are set in bold italic typeface, e.g. *Press the **SCALE** Key*.

Chapter Layout

Introduction — Lists and describes the components of the kit. Also identifies the parts of the DPM, describes the functions of the keys, and explains the meaning of indicators which may be displayed.

Setup — Gives directions for setting up the kit and connecting the DPM, DPS, and load.

Operation — Explains how to make measurements with the DPM, and the special functions used with specific sensors.

Maintenance — Lists routine maintenance tasks for the kit, and troubleshooting tips for common problems. Specifications and battery information are also included.

Changes to this Manual

We have made every effort to ensure this manual is accurate at the time of publication. If you should discover any errors or if you have suggestions for improving this manual, please send your comments to our factory. This manual may be periodically updated, when inquiring about updates to this manual refer to the part number and revision level on the title page.

Table of Contents

Chapter 1 Introduction

Items Supplied

The 3T Power Measurement Kit has the following components:

Carrying Case — Hard case for storage of the kit. Wheeled for easier transport.

Bird 5000-EX (DPM) — Digital power meter used with the DPS. Has internal batteries, can also be powered from ac adapter. Can display average or peak power, forward and reflected power, and return loss.

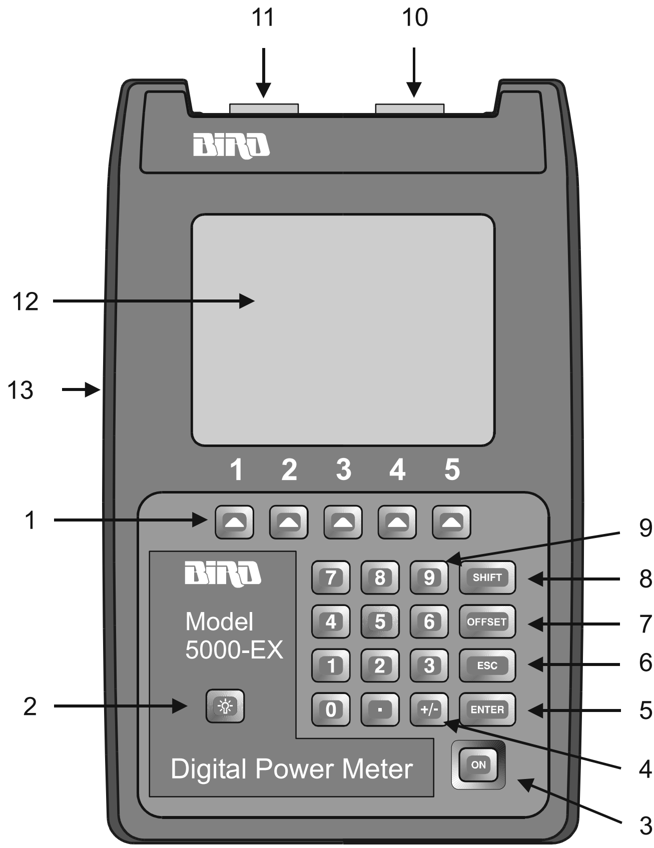
Bird 5010G (DPS) — Directional power sensor used with the DPM. Thruline measurement. Bird 43-Type Elements are provided. Can also use Bird APM-Type elements.

Bird 8581A Air Load — Air-cooled load. The load has no controls, settings, or moving parts.

Assorted elements, cables, and adapters are also included in the kit (See "Parts List" on page 16 for an exploded view and a complete list).

DPM Keys and Connections

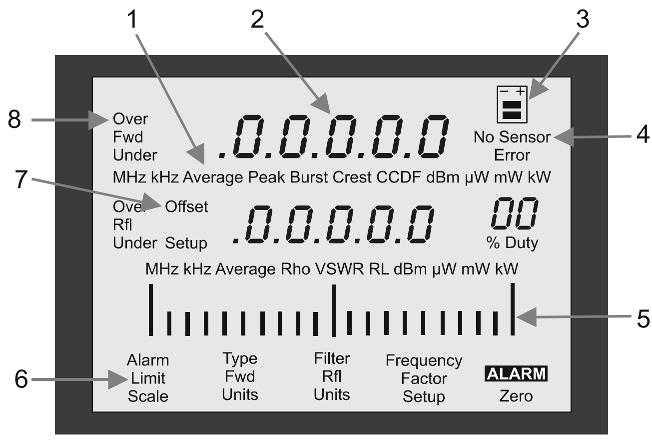
Figure 1
DPM Keys



1. Speed Keys	Activates the command displayed above it.
2. Backlight Key	Turns the backlight on momentarily.
3. On Key	Turns the DPM on or off.
4. +/- Key	Toggles between positive and negative numbers. Toggles between the first and second set of speed key functions.
5. Enter Key	Completes data entry.
6. Escape Key	Aborts data entry without accepting changes.
7. Offset Key	Enter offset values in dB.
8. Shift Key	Toggles between different sets of speed key functions.
9. Numeric Keys	Input numeric values.
10. Sensor Port	Connection for power sensors.
11. RS-232 Port	Connection for an optional PC 9-pin RS-232 (DB9) connector, compatible with PC serial port. 9600 baud, 8 data bits, 1 stop bit, and no parity.
12. LCD Display	Backlit liquid crystal display.
13. External DC Connector	Connection for the ac adapter. External supplies power the unit and charge the internal battery.

DPM Display

Figure 2
DPM Display



1. Display Units	Shows the measurement mode and units for the display directly above.
2. Display	The DPM has two measurement displays. The top displays forward average power, peak power, or the average of the max. and min. peaks. The bottom displays reflected power and reflection measurement (VSWR).
3. Battery Level Indicator	When the external adapter is connected, the indicator blinks until the battery is fully charged. When using the battery, the indicator is on continuously and the black bars show the battery change remaining. When the battery is low, the indicator will be empty and will blink. Recharge the batteries.
4. No Sensor Indicator	Turns on when no sensor is connected to the sensor port.
5. Analog Bar Graph	Proportional to the forward average power. Minimum at 0 W or –20 dBm. Max at full scale power.

6. **Speed Key Labels** Press the speed key below the label to activate the displayed function.

7. **Offset Indicator** Blinks when an offset is in use.

8. **Power Indicator** If the power exceeds 100% of full scale, “Over” will be displayed. If the power is less than zero, “Under” will be displayed. If the power exceeds 120% of full scale, the display will show a line of dashes and the bar scale will be filled.

Speed Keys

Set 1 These speed keys are initially displayed when a DPS is first connected. Press **SHIFT** to switch to Set 2.

***Scale* —**Sets the forward full scale power. This is listed on the forward element’s nameplate. Reflected full scale power is automatically set to 10 dB below the forward full scale.

***Fwd Units* —**Selects the units for the major display.

***Rfl Units* —**Selects the units for the minor display.

Set 2 Press **SHIFT** to switch to Set 1.

***Type* —**Sets the power measurement mode. Bird 43 Elements can measure average, peak, and average of max. and min. power. Bird APM Elements can only measure average power.

Setup* —**Sets the element type. Press ***Type to switch between 43 (default) and APM elements. Press **ENTER** when done.

To set up the kit to make measurements, follow the steps below. They are described in detail later in this chapter.

1. Turn off the signal source.
2. Connect the source to the power sensor (DPS), the DPS to the load, and the DPS to the meter (DPM).
3. Put the appropriate elements in the DPS.
4. Turn on the DPM and set the full scale power.

Power Supply

The Bird DPM uses a rechargeable Nickel-Metal Hydride battery pack. Charge life is a minimum of 100 hours continuous usage. “Low Bat” is displayed when the batteries require charging.

☞ NOTE: For optimum battery life, charge the batteries only after “Low Bat” is displayed.

The DPM can use an external power source. Using the DPM with the ac adapter will also charge the battery. Charging time is 8 hours from full discharge.

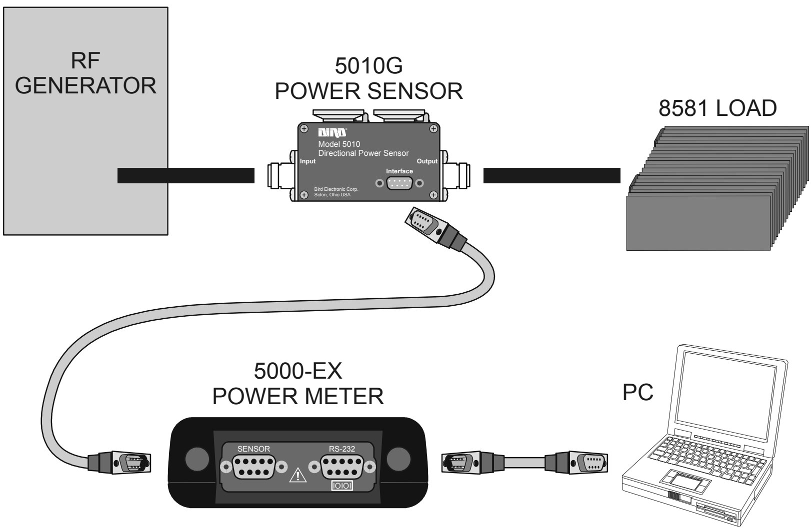
> CAUTION
> Use a Bird adapter only and do not use the adapter with the batteries removed.

To use the ac adapter, insert the adapter’s barrel connector into the DPM’s external dc connector (See Figure 1 on page 2). Insert the adapter plug into a wall receptacle.

☞ NOTE: The DPM batteries are shipped uncharged for safety. Charge the batteries before using.

Connections

Figure 3
DPM Connections

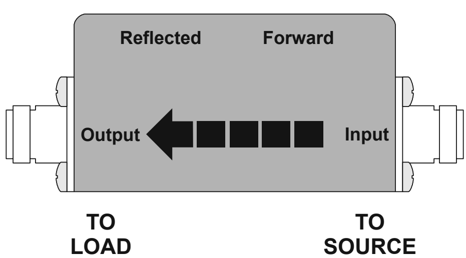


> WARNING
> Never attempt to connect or disconnect RF equipment from the transmission line while RF power is being applied.
> Leaking RF energy is a potential health hazard.

Connecting to RF Power

1. Connect the DPS Output to the load using the 7/16 to 7/16 cable provided. The arrow on the DPS should point towards the load (see Figure 4 below).
2. Use the supplied adapters and cables to connect the DPS Input to the RF Source.

Figure 4
DPS RF Connections



Connecting to the DPM

> CAUTION
> Always turn off the DPM before connecting or disconnecting a sensor.

Although unlikely, it is possible to corrupt the power sensor firmware by connecting it to the DPM while the DPM is on. To prevent this, always turn the DPM off before connecting or disconnecting a sensor.

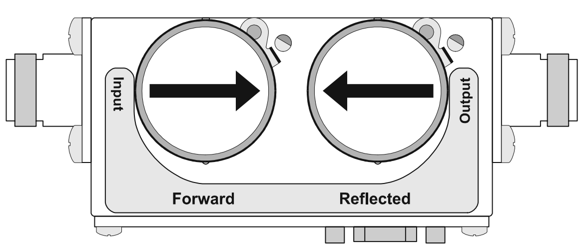
Connect the power sensor to the “Sensor” port on the DPM using the sensor cable provided. If you are using the optional PCTool software, connect the PC’s serial port to the “RS-232” port on the DPM.

Elements

The Directional Power Sensor (DPS) uses Bird Plug-In Elements. These are labeled with a max power and a frequency range. The generator frequency should be within the element range, and the forward element power should be 10x the reflected element power.

The element in the forward socket should be inserted with its arrow pointing in the direction of forward power, as shown below. The reflected element should point in the direction of reverse power.

Figure 5 Element Orientation



Setting Forward Scale

For the DPS, the forward full scale power and element type must be entered manually. The reflected full scale power is automatically set to 1/10 of the forward full scale.

- Press ***SCALE***.
- Change the units with the ***FWD Units*** key until the DPM units match the units on the forward element.
- Use the numeric keypad to enter the maximum power of the element in the forward element socket.

☞ NOTE: The element's max power is listed on the element nameplate.

- Press **ENTER**.

Changing Element Type

Bird 43-Type elements are included with the kit. To use Bird APM-Type elements, the DPM will need to change modes:

- Press **SHIFT**.
- Press ***Setup***.
- Change the element type with the ***Type*** key until the display agrees with the description on the element nameplate. ***A*** is displayed for APM type elements. ***43*** is displayed for 43 Type elements.
- Press **ENTER**.
- Press **SHIFT**.

After the kit has been set up, the DPM turned on, and the element settings have been entered, the DPM is ready to take measurements.

- Select the measurement by pressing **SHIFT**, then pressing ***Type*** until the correct measurement is shown. Press **SHIFT** to return to the main menu.
- Set the measurement units by pressing ***FWD Units*** and ***RFL Units***.
- If you know the system loss or are using an attenuator, add the losses (in dB) of all components in the system. For attenuators and other frequency-dependent components, use the loss at the measured frequency. Then, press **OFFSET** and enter the total loss in dB. This will allow you to read the actual line power. The DPM accepts offsets from 0 to 100 dB.
- Turn on the RF source.
- Read the power level or other measurement on the display.

☞ NOTE: The bar graph will respond immediately to changes in the RF power. The major and minor displays will respond after a one second delay.

- When done, turn off the RF source.
- Turn off the DPM.

WARNING Load may be hot. Allow load to cool before touching.

CAUTION Packing foam may melt if heated. Allow load to cool before packing.

- Wait 20 minutes for the load to cool to room temperature.
- Put kit components back into case.

Troubleshooting

PROBLEM	**POSSIBLE CAUSE**	**REMEDY**
Nothing shown on display	Unit is off	Momentarily press the **ON** key.
	Batteries are drained	Use external power source.
		Replace the batteries.
Display shows dashes and “Over”; bar scale is full	Unit is overrange	Use higher power elements or reduce RF power.
Erratic power readings	Element contact out of alignment	Align the contact. It must be far enough out to make good contact, but must not restrict entry of the element body.
	Damaged element	Replace element.
	Sensor is damaged	Replace sensor.

Charging Batteries

Fully charged batteries provide a minimum of 100 hours of continuous operation. Charging time is typically 8 hours using the ac adapter. The batteries charge whenever the DPM is connected to ac power sources with the ac Mains adapter. The unit will charge with its power turned either on or off.

☞ NOTE: For optimum battery life, charge the batteries only after “Low Bat” is displayed.

Cleaning

CAUTION Harsh or abrasive detergents and some solvents can damage the display unit and information on the labels.

Clean the Bird Digital Power Meter and its display with a soft cloth dampened with mild detergent and water only. Clean sensors with a dry cleaning solvent that leaves no residue.

Battery Replacement

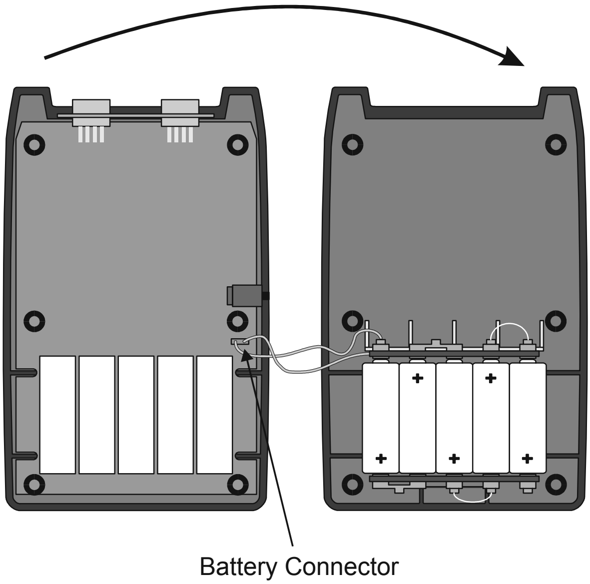
The Nickel-Metal Hydride (Ni-MH) batteries do not normally need replaced. If necessary however, follow these instructions (see Figure 6 below).

WARNING Disconnect from external power before any disassembly. The potential for electric shock exists.

CAUTION Replace only with Ni-MH rechargeable A batteries. Nominal Voltage 1.2V; Capacity 2700mAhr.

- Lay the DPM, display side down, on a clean surface.
- Use a small screwdriver to remove all six screws from the back cover.
- Taking care to not disconnect the battery connector, lift the back cover off, flip it over, and lay it, battery side up, next to the front cover.
- Remove the old batteries.
- Install the new batteries checking the orientation of the positive and negative terminals.
- Be sure the battery connector is connected.
- Taking care to not pinch the battery connector cables, place the back cover onto the front and make sure it is properly seated.
- Replace the screws.

Figure 6
Back Cover
Removal



Customer Service

Any maintenance or service procedure beyond the scope of those in this chapter should be referred to a qualified service center.

If you need to return the unit for any reason, contact the Bird Service Center for a return authorization. All instruments returned must be shipped prepaid and to the attention of Bird Service Center.

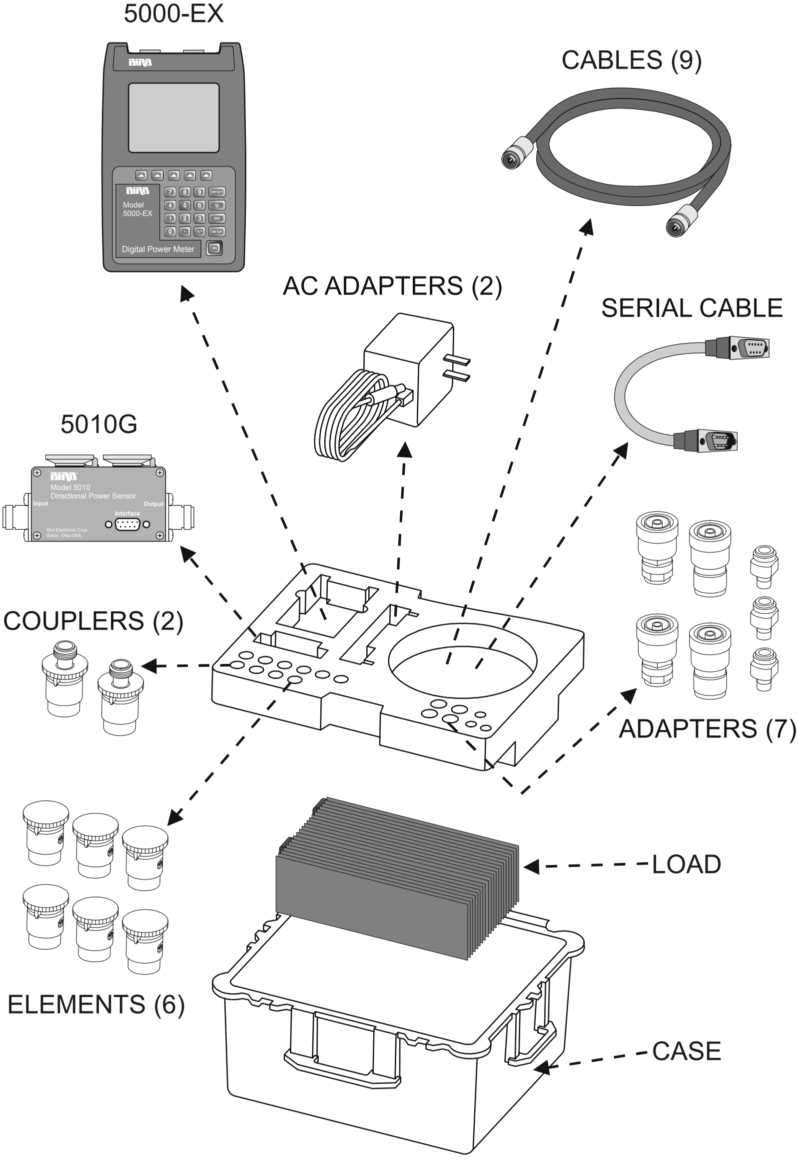
Bird Service Center
30303 Aurora Road
Cleveland (Solon), Ohio 44139-2794
Phone: (440) 519-2298
Fax: (440) 519-2326
E-mail: *bsc@bird-technologies.com*

For the location of the Sales Office nearest you, give us a call or visit our Web site at:

http://www.bird-electronic.com

Parts List

Figure 7
Supplied Items



Part Name	Bird P/N	GE P/N
Kit, Complete	GE3TKIT	2372868
Carrying Case	4300A672-1	2372868-2
Digital Power Meter	5000-EX	2372868-3
AC Adapters		2372868-4
115 Vac	5A2229	
230 Vac	5A2226	
Directional Power Sensor, Elements, and Offset Card Kit	4300A680	2372868-5
9-pin Serial Cable	5A2264-09-MF-10	2372868-6
Cables		
Two 7/16 DIN (M) – 7/16 DIN (M), 6 ft	4300A678-8	2372868-7
BNC (M) – BNC (M), 6 ft	4300A676-6	2372868-8
BNC (M) – SMB (M), 4 ft	4300A676-5	2372868-9
BNC (M) – SMB (F), 4 ft	4300A676-9	2372868-10
BNC (M) – SMA (F), 4 ft	4300A676-4	2372868-11
BNC (M) – SMA (M), 4 ft	4300A676-3	2372868-12
BNC (M) – Phoenix (M), 4 ft	4300A676-2	2372868-13
BNC (M) – Phoenix (F), 4 ft	4300A676-1	2372868-14
Adapters		
7/16 DIN(F) – HN(M)	4240-500-7	2372868-15
7/16 DIN(F) – N(M)	PA-MNFE	2372868-16
7/16 DIN(M) – HN(F)	4240-500-9	2372868-17
7/16 DIN(M) – N(F)	PA-FNME	2372868-18
N(F) – BNC(F)	4240-500-8	2372868-19
Two BNC(F) – BNC(F)	4240-500-20	2372868-20
Dust Plug	4230-022	2372868-21
Coupling Element		
60 dB	4300A375-1	2372868-22
40 dB	4300A375-2	2372868-23
Load, 2 kW Air-Cooled	8581A230-1	2372868-24

Color Codes

Sensor Elements

Frequency Range	Color
10 – 30 MHz	Black
30 – 60 MHz	Green
60 – 130 MHz	Blue

Coupler Elements

Coupling Factor	Color
40 dB	Red
60 dB	Brown

Cables

GE P/N	Color
2372868-7	Red
2372868-8	Orange
2372868-9	Yellow
2372868-10	Green
2372868-11	Blue
2372868-12	Violet
2372868-13	Blue/Green
2372868-14	Red/Yellow

Specifications

Overall Kit

Power, Max. Pulse	37 kW
Pulse Width Duty Cycle	3.2 ms (4 ms max.) 6% duty cycle
Frequency Range	10 – 130 MHz
Accuracy	± 3% at calibrated frequency/power levels ± 8% otherwise
Connector Type	7/16 DIN (F)
Meter	5000-EX
Sensor	5010G
Load	8581A
VSWR Stability	1.1:1 ± 0.1 dB over 10 minutes
Impedance	50 Ohm
Operating Position (Load Only)	Vertical
Case	Rugged case with multiple handles and two wheels for portability
Dimensions, Case	24.75” x 19.5” x 14” (630 x 492 x 352 mm)
Calibration Cycle	6 months
Weight, Nominal	< 65 lbs. (30 kg) Fully Assembled
Material of Construction	No toxic materials, sharp edges, or projections that present environmental or personnel hazards.
Magnetic Field	500 Gauss
Altitude*	
Operating Storage	–984 to +11,800 feet above sea level –984 to +18,000 feet above sea level

Temperature Operating Storage	 5 to 40 °C (41 to 104 °F) –40 to +70 °F (–40 to +158 °C)
Humidity, Relative Operating Storage	 30% to 75% 5% to 100%

* Derate power above 5,000 feet. See Figure 8 on page 22.

Bird 5000-EX Digital Power Meter

Display	Backlit LCD with major and minor displays and analog bar graph
Major Display	5 digits, 5/16” High
Minor Display	5 digits, 1/4” High
Analog Bar Graph	20 segment, horizontal orientation. Tracks the major digital display
Measurement Modes	Average power, peak power, or average peak power depending on the element or sensor.
Sensor Interface	9-pin D-shell RS-232 serial connector. Sensor is powered from the DPM.
PC Interface	9-pin RS-232 serial port
Power Supply Battery AC	 Rechargeable Nickel-Metal Hydride (NiMH) Meter may be operated from ac mains using supplied adapter. 120 ± 10% Vac @ 50/60 ± 3 Hz
Battery Life	100 hours, minimum, per charge.
Battery Charge Time	8 hours for full charge.
Dimensions	8.0”H x 4.625”W x 2.0”D (203 x118 x 51 mm)
Weight	2 lbs. (0.9 kg) nominal

Bird 5010G Directional Power Sensor	
Frequency Range	10 – 130 MHz
Accuracy	± 3% at calibrated frequency/power levels ± 8% otherwise
Dynamic Range	20 dB
Sensor Type	Thruline directional dual-element line section.
Elements*	Can use Bird 43 or APM Elements. 43 elements are standard
Settling Time	< 3 seconds
Insertion VSWR	1.05:1
Dimensions	2.5”H x 5.0”W x 2.0”D (58 x 127 x 51 mm)
Weight	1.25 lb. (0.57 kg) nominal with elements.

* RFL element power must be 1/10 of FWD power.

Coupling Elements		
Connector	BNC(F)	
Coupling	60 – 130 MHz	10 – 60 MHz
60 dB	–60 ± 0.5 dB	–60 ± 2.5 dB
40 dB	–40 ± 0.5 dB	–48 dB max.

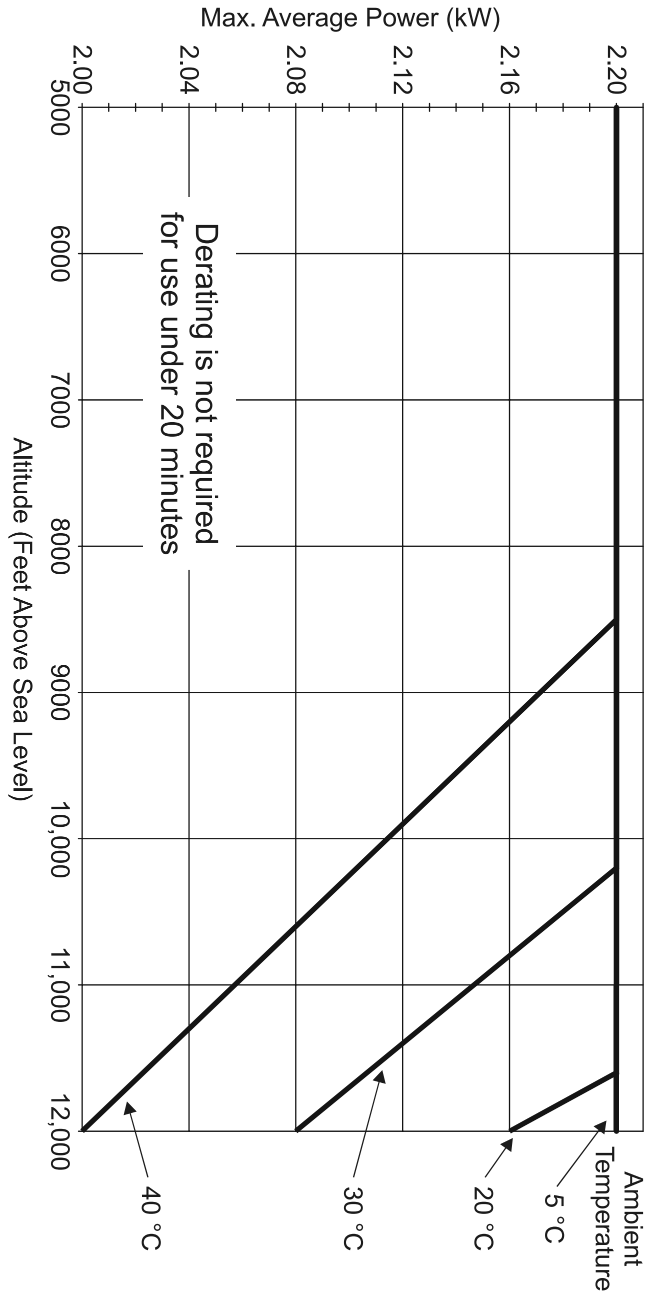


Figure 8
High Altitude Power Derating

Limited Warranty

All products manufactured by Seller are warranted to be free from defects in material and workmanship for a period of one (1) year, unless otherwise specified, from date of shipment and to conform to applicable specifications, drawings, blueprints and/or samples. Seller's sole obligation under these warranties shall be to issue credit, repair or replace any item or part thereof which is proved to be other than as warranted; no allowance shall be made for any labor charges of Buyer for replacement of parts, adjustment or repairs, or any other work, unless such charges are authorized in advance by Seller.

If Seller's products are claimed to be defective in material or workmanship or not to conform to specifications, drawings, blueprints and/or samples, Seller shall, upon prompt notice thereof, either examine the products where they are located or issue shipping instructions for return to Seller (transportation charges prepaid by Buyer). In the event any of our products are proved to be other than as warranted, transportation costs (cheapest way) to and from Seller's plant, will be borne by Seller and reimbursement or credit will be made for amounts so expended by Buyer. Every such claim for breach of these warranties shall be deemed to be waived by Buyer unless made in writing within ten (10) days from the date of discovery of the defect.

The above warranties shall not extend to any products or parts thereof which have been subjected to any misuse or neglect, damaged by accident, rendered defective by reason of improper installation or by the performance of repairs or alterations outside of our plant, and shall not apply to any goods or parts thereof furnished by Buyer or acquired from others at Buyer's request and/or to Buyer's specifications. Routine (regularly required) calibration is not covered under this limited warranty. In addition, Seller's warranties do not extend to the failure of tubes, transistors, fuses and batteries, or to other equipment and parts manufactured by others except to the extent of the original manufacturer's warranty to Seller.

The obligations under the foregoing warranties are limited to the precise terms thereof. These warranties provide exclusive remedies, expressly in lieu of all other remedies including claims for special or consequential damages. SELLER NEITHER MAKES NOR ASSUMES ANY OTHER WARRANTY WHATSOEVER, WHETHER EXPRESS, STATUTORY, OR IMPLIED, INCLUDING WARRANTIES OF MERCHANTABILITY AND FITNESS, AND NO PERSON IS AUTHORIZED TO ASSUME FOR SELLER ANY OBLIGATION OR LIABILITY NOT STRICTLY IN ACCORDANCE WITH THE FOREGOING.